Python Questions on Tuple and Set

In [1]:
```python
# What is a tuple in Python and how is it different from a list?

# Tuple(): It is a collection which is ordered, indexed and unchangeable, allows duplicate members.
# Whereas lists() is also a collection which is odered, indexed and changeable, allows duplicate members.
# The only difference is lists() are changeable and tuples() are not changeable/immutable.
```

In [4]:
```python
# How can you create a tuple with a single element?
mtp = ('Innomatics', )
print(type(mtp))
# By placing a comma(",") after the single elment it will consider as tuple other wise it will consider as string.
```

```
<class 'tuple'>
```

In [7]:
```python
# Is a tuple mutable or immutable? Justify your answer.

# Tuples() are immutabble:
mtp1 = 'my', 'name', 'is', 'Yaswanth'
print(mtp1[2])
print(mtp1[2] = IS)
# As you see the above example it is giving a error because tuples() are immutable.
```

```
  Cell In[7], line 6
    print(mtp1[2] = IS)
          ^
SyntaxError: expression cannot contain assignment, perhaps you meant "=="?
```

In [8]:
```python
# Write a Python code to convert a list into a tuple.
mlt1 = ['my', 'name', 'is', 'Yaswanth']
print(tuple(mlt1))
```

```
('my', 'name', 'is', 'Yaswanth')
```

In [9]:
```python
# How do you access elements in a tuple?
mtp1 = 'my', 'name', 'is', 'Yaswanth'
print(mtp1[2])
print(mtp1[2:5])
print(mtp1[1:5:2])
print(mtp1[::-1])
```

```
is
('is', 'Yaswanth')
('name', 'Yaswanth')
('Yaswanth', 'is', 'name', 'my')
```

In [10]:
```python
# What happens if you try to change the value of a tuple element?

# It will give a Error immedietly:
mtp1 = 'my', 'name', 'is', 'Yaswanth'
mtp1[0] = 'MY'
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[10], line 5
      1 # What happens if you try to change the value of a tuple element?
      2 
      3 # It will give a Error immedietly:
      4 mtp1 = 'my', 'name', 'is', 'Yaswanth'
----> 5 mtp1[0] = 'MY'

TypeError: 'tuple' object does not support item assignment
```

In [18]:
```python
# Can a tuple contain elements of different data types? Explain with example.
# Yes, tuples() are hetrogenous.
mtp2 = 'Innomatics', 23, 46.43, True
print(mtp2)
```

```
('Innomatics', 23, 46.43, True)
```

In [13]:
```python
# How do you check if an item exists in a tuple?
# By membership operator "in"
mtp2 = 'Innomatics', 23, 46.43, True
print(23 in mtp2)
```

```
True
```

In [14]:
```python
# Write a program to count the number of occurrences of an element in a tuple.
mtp1 = 'my', 'name', 'is', 'Yaswanth', 'is'
print(mtp1.count("is"))
```

```
2
```

In [19]:
```python
# How do you find the index of an element in a tuple?
mtp1 = 'my', 'name', 'is', 'Yaswanth', 'is'
print(mtp1.index("Yaswanth"))
```

```
3
```

In [31]:
```python
# Explain slicing in tuples with an example.
mtp1 = 'my', 'name', 'is', 'Yaswanth', 'is'
print(tuple(enumerate(mtp1)))
print(mtp1[3])
print(mtp1[3][0:5])
print(mtp1[3][0:5][1:6:2])
```

```
((0, 'my'), (1, 'name'), (2, 'is'), (3, 'Yaswanth'), (4, 'is'))
Yaswanth
Yaswa
aw
```

In [35]:
```python
# Can a tuple contain another tuple? Give an example.
mtp3 = 'my', 'name', 'is', 'Yaswanth', 'is', ('Innomatics', 23, 46.43, True)
print(mtp3, type(mtp3))
print(len(mtp3))
```

```
('my', 'name', 'is', 'Yaswanth', 'is', ('Innomatics', 23, 46.43, True)) <class 'tuple'>
6
```

In [126…
```python
# Write a code to iterate over a tuple using a for loop.
mtp3 = 'my', 'name', 'is', 'Yaswanth', 'is', ('Innomatics', 23, 46.43, True)
for char in mtp3:
    print(char)
```

```
my
name
is
Yaswanth
is
('Innomatics', 23, 46.43, True)
```

In [38]:
```python
# What are the built-in methods available for tuples?
print(dir(tuple))
# These are tuple() built-in methods avaliable for tuples()
```

```
['__add__', '__class__', '__class_getitem__', '__contains__', '__delattr__', '__dir__', '__doc__', '__eq__', '__format__', '__ge__', '__getattribute__', '__getitem__', '__getnewargs__', '__ge
tstate__', '__gt__', '__hash__', '__init__', '__init_subclass__', '__iter__', '__le__', '__len__', '__lt__', '__mul__', '__ne__', '__new__', '__reduce__', '__reduce_ex__', '__repr__', '__rmul
__', '__setattr__', '__sizeof__', '__str__', '__subclasshook__', 'count', 'index']
```

In [41]:
```python
# Demonstrate the use of the `count()` method with tuples.
mtp3 = 'my', 'name', 'is', 'Yaswanth', 'is', ('Innomatics', 23, 46.43, True), 'is', 'orange'
print(mtp3.count("is"))
```

```
3
```

In [42]:
```python
# Demonstrate the use of the `index()` method with tuples.
mtp3 = 'my', 'name', 'is', 'Yaswanth', 'is', ('Innomatics', 23, 46.43, True), 'is', 'orange'
print(mtp3.index("orange"))
```

```
7
```

In [124…
```python
# Can a tuple be used as a dictionary key? Why or why not?

# Yes tuples can be used as dictionary key, when they are immutable.
# Dictionary keys must be hashable, the value cannot change.
# So, if tuple contains only immutable value types like strings,numbers,other strings, then these can be used as dictionary keys.
# If tuple contains any mutable objects like list, they are not hashable and cannot be used as dictionary keys.
# Example:
dt1 = {(1 , 2): "value"}  # Valid
dict2 = {([1 , 2]): "value"} # Not Valid
# So tuples with immutable objects only can be used as dictionary keys.
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[124], line 9
      1 # Can a tuple be used as a dictionary key? Why or why not?
      2 
      3 # Yes tuples can be used as dictionary key, when they are immutable.
   (...)
      6 # If tuple contains any mutable objects like list, they are not hashable and cannot be used as dictionary keys.
      7 # Example:
      8 dt1 = {(1 , 2): "value"}  # Valid
----> 9 dict2 = {([1 , 2]): "value"}

TypeError: unhashable type: 'list'
```

In [46]:
```python
# How do you unpack values from a tuple into variables?
mtp3 = 'my', 'name', 'is', 'Yaswanth', 'is', ('Innomatics', 23, 46.43, True), 'is', 'orange'
a,b,c,d,e,f,g,h = mtp3
print(f, "\n",d )
```

```
('Innomatics', 23, 46.43, True)
 Yaswanth
```

In [47]:
```python
# What is tuple packing and unpacking?
# Tuple packing: Means grouping multiple values into a single tuple()
# We can assign multiple elements/values consisting of different data types, python will consider as a tuple by simply giving in tuple syntax form.

# Tuple Unpacking: Extracting the values form a tuple into seperate variables.
# The number of variables must be equal to values inside the tuple.
```

In [50]:
```python
# Compare tuples and lists in terms of performance.
# List():
mlt2 = ['my', 'name', 'is', 'Yaswanth']
mlt2[0] = 'MY'
print(mlt2)

mtp1 = 'my', 'name', 'is', 'Yaswanth'
mtp1[0] = 'MY'
# As we see the above example the list is able to change the value or modify it but tuple is giving error.
# Performance wise both are as same as they are except mutablility.
```

```
['MY', 'name', 'is', 'Yaswanth']
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[50], line 8
      5 print(mlt2)
      7 mtp1 = 'my', 'name', 'is', 'Yaswanth'
----> 8 mtp1[0] = 'MY'

TypeError: 'tuple' object does not support item assignment
```

In [54]:
```python
# Create a nested tuple and access its elements.
mtp3 = 'my', 'name', 'is', 'Yaswanth', 'is', ('Innomatics', 23, 46.43, True), 'is', 'orange'
print(tuple(enumerate(mtp3)))
print(mtp3[5])
print(mtp3[5][2])
```

```
((0, 'my'), (1, 'name'), (2, 'is'), (3, 'Yaswanth'), (4, 'is'), (5, ('Innomatics', 23, 46.43, True)), (6, 'is'), (7, 'orange'))
('Innomatics', 23, 46.43, True)
46.43
```

In [55]:
```python
# How can you find the length of a tuple?
mtp3 = 'my', 'name', 'is', 'Yaswanth', 'is', ('Innomatics', 23, 46.43, True), 'is', 'orange'
print(len(mtp3))
```

```
8
```

In [57]:
```python
# What is the use of the `in` operator with tuples?
mtp3 = 'my', 'name', 'is', 'Yaswanth', 'is', ('Innomatics', 23, 46.43, True), 'is', 'orange'
print(True in mtp3[5])
```

```
True
```

In [60]:
```python
# How do you concatenate two tuples?
mtp4 = 'Innomatics', 'Institute'
mtp5 = 'is', 'excellent'
print(mtp4+mtp5)
```

```
('Innomatics', 'Institute', 'is', 'excellent')
```

In [59]:
```python
# Explain tuple immutability with an example.
mtp1 = 'my', 'name', 'is', 'Yaswanth'
mtp1[0] = 'MY'
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[59], line 3
      1 # Explain tuple immutability with an example.
      2 mtp1 = 'my', 'name', 'is', 'Yaswanth'
----> 3 mtp1[0] = 'MY'

TypeError: 'tuple' object does not support item assignment
```

In [61]:
```python
mtp3 = 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange'
print(mtp3*6)
```

```
('my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange', 'my', 'name', 'is',
'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'In
nomatics', 23, 46.43, True, 'is', 'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange')
```

In [63]:
```python
# What is a set in Python and how is it different from a list and a tuple?
st1 = {'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is',
       'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange',
       'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange', 'my', 'name', 'is',
       'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics',
       23, 46.43, True, 'is', 'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange'}
print(st1)
# As you can see the difference, it do not allows duplicate members, and does't follow any user order it takes the position of values randomly.
# sets are unordered, unindexed, immutable, also no duplicates are allowed.
```

```
{True, 'Yaswanth', 'Innomatics', 'is', 46.43, 'my', 23, 'orange', 'name'}
```

In [65]:
```python
# How do you create a set?
st1 = {'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is',
       'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange',
       'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange', 'my', 'name', 'is',
       'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics',
       23, 46.43, True, 'is', 'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange'}
print(st1)
# With curly braces a set can be created.
```

```
{True, 'Yaswanth', 'Innomatics', 'is', 46.43, 'my', 23, 'orange', 'name'}
```

In [66]:
```python
# Can a set have duplicate values? Explain.

# No, a set doesn't allow duplicate values as it was designed in that way.
# Example:
st1 = {'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is',
       'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange',
       'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange', 'my', 'name', 'is',
       'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics',
       23, 46.43, True, 'is', 'orange', 'my', 'name', 'is', 'Yaswanth', 'is', 'Innomatics', 23, 46.43, True, 'is', 'orange'}
print(st1)
```

```
{True, 'Yaswanth', 'Innomatics', 'is', 46.43, 'my', 23, 'orange', 'name'}
```

In [67]:
```python
# How do you add an element to a set?
my_set = {56,243,367}
my_set.add(50235)
print(my_set)
```

```
{56, 50235, 243, 367}
```

In [68]:
```python
# How do you remove an element from a set?
my_set.remove(50235)
print(my_set)
```

```
{56, 243, 367}
```

In [73]:
```python
# What is the difference between `remove()` and `discard()` in sets?
my_set = {56, 50235, 243, 367}
print(my_set.discard(502357))

print("--"*46)

my_set = {56, 50235, 243, 367}
print(my_set.remove(502353))

# As you can see that when you try to eleminate a non exsisting element from the set using .remove(), it will give error.
# Where as when you try that using .discard(), it will give a output as "None" if this was the case or else
```

```
None
--------------------------------------------------------------------------------------------
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[73], line 8
      5 print("--"*46)
      7 my_set = {56, 50235, 243, 367}
----> 8 print(my_set.remove(502353))

KeyError: 502353
```

In [74]:
```python
# What is the use of `pop()` in sets?
my_set.pop()
print(my_set)
# pop(): Removes and returns an arbitrary element from the set.
# Raises KeyError if the set is empty.
```

```
{50235, 243, 367}
```

In [76]:
```python
# How do you clear all elements from a set?
my_set = {56, 50235, 243, 367}
my_set.clear()
print(my_set)
```

```
set()
```

In [132…
```python
# What are set operations in Python?
# 1) Union, 2)Intersection
mst =  {True, 'Yaswanth', 'Innomatics', 'is', 46.43, 'my', 23, 'orange', 'name', 56}
mst1 = {56, 50235, 243, 367}
print(mst.union(mst1))
print("--"*30)
print(mst.intersection(mst1))
```

```
{True, 'Yaswanth', 'Innomatics', 'is', 46.43, 367, 'my', 243, 23, 56, 'orange', 'name', 50235}
------------------------------------------------------------
{56}
```

In [135…
```python
mst =  {True, 'Yaswanth', 'Innomatics', 'is', 46.43, 'my', 23, 'orange', 'name', 56}
mst1 = {56, 50235, 243, 367}
print(mst.union(mst1))
print("--"*30)
print(mst.intersection(mst1))
```

```
{True, 'Yaswanth', 'Innomatics', 'is', 46.43, 367, 'my', 243, 23, 56, 'orange', 'name', 50235}
------------------------------------------------------------
{56}
```

In [138…
```python
# How do you find the difference between two sets?
mst =  {True, 'Yaswanth', 'Innomatics', 'is', 46.43, 'my', 23, 'orange', 'name', 56}
mst1 = {56, 50235, 243, 367}
print(mst.difference(mst1))
print(mst-mst1) # Second method
```

```
{True, 'Yaswanth', 'Innomatics', 'is', 46.43, 'my', 23, 'orange', 'name'}
{True, 'Yaswanth', 'Innomatics', 'is', 46.43, 'my', 23, 'orange', 'name'}
```

In [140…
```python
# How do you find the symmetric difference between sets?
mst =  {True, 'Yaswanth', 'Innomatics', 'is', 46.43, 'my', 23, 'orange', 'name', 56}
mst1 = {56, 50235, 243, 367}
print(mst.symmetric_difference(mst1))
print(mst^mst1) # Second method
```

```
{True, 'Yaswanth', 'Innomatics', 'is', 46.43, 367, 'my', 243, 23, 'orange', 'name', 50235}
{True, 'Yaswanth', 'Innomatics', 'is', 46.43, 367, 'my', 243, 23, 'orange', 'name', 50235}
```

In [142…
```python
# Explain subset and superset with code examples.
sb = {1 , 2 , 3}
sb1 = {1 , 2}
print(sb.issubset(sb1))
print(sb.issuperset(sb1))
```

```
False
True
```

In [145…
```python
# How do you check if two sets are disjoint?
mst =  {True, 'Yaswanth', 'Innomatics', 'is', 46.43, 'my', 23, 'orange', 'name', 56}
mst1 = {56, 50235, 243, 367}
print(mst.isdisjoint(mst1))

sb = {1 , 2 , 3}
sb1 = {4 , 5}
print(sb.isdisjoint(sb1))
```

```
False
True
```

In [92]:
```python
# Can you add a list to a set? Why or why not?

# No, you cannot add a list directly to a set in Python because lists are mutable, and sets only contain immutable elements.
# If you try to do this, you will get a TypeError.
```

Python Questions on Tuple and Set

In [123…
```python
# How do you convert a list to a set?
my_list = [56, 50235, 243, 367]
my_set2 = set(my_list)
my_set2
```

Out[123…
```
{56, 243, 367, 50235}
```

In [94]:
```python
# How do you find the length of a set?
my_set = {56, 50235, 243, 367}
print(len(my_set))
```

```
4
```

In [100…
```python
# Can a set contain mixed data types? Give example.
# Yes a set can contain mixed data types but, We cannot perform operations with having different data types.
my_set = {56, 50235, 243, 367, 'Soap', 54.767}
my_set
```

Out[100…
```
{243, 367, 50235, 54.767, 56, 'Soap'}
```

In [101…
```python
# How do sets help in removing duplicates from a list?

# Sets automatically removes duplicates because they only store unique elements.
# By converting a list to a set, duplicates are removed, and you can convert it back to a list for unique elements.
```

In [102…
```python
# What will happen if you try to access set elements using index?
my_set = {56, 50235, 243, 367, 'Soap', 54.767}
print(my_set[3])
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[102], line 3
      1 # What will happen if you try to access set elements using index?
      2 my_set = {56, 50235, 243, 367, 'Soap', 54.767}
----> 3 print(my_set[3])

TypeError: 'set' object is not subscriptable
```

In [103…
```python
# Demonstrate a scenario where sets can improve performance.
```

In [152…
```python
# What are frozen sets and how are they different from regular sets?

sb = {1 , 2 , 3}
sb1 = {5 , 6}
sb.add(4)
print(sb)

print("--"*30)

fsb = frozenset(sb1)
fsb.add(7)
print(fsb)

# The difference is regular sets are able to use methods like add(), some of methods may not work.
```

```
{1, 2, 3, 4}
------------------------------------------------------------
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[152], line 11
      8 print("--"*30)
     10 fsb = frozenset(sb1)
---> 11 fsb.add(7)
     12 print(fsb)

AttributeError: 'frozenset' object has no attribute 'add'
```

In [154…
```python
# How do you create a frozen set?

# By declaring it to a new variable and converting set to frozen set, Example:
sb1 = {5 , 6}
fsb = frozenset(sb1)
fsb
```

Out[154…
```
frozenset({5, 6})
```

In [156…
```python
# Can a set be nested? Explain.

s={5,3,2,{'dress','hat'},7}
print(s)

# 1. Sets cannot be nested directly because their elements must be immutable,
# and regular sets are mutable.

# 2. You can include frozen sets which are immutable,
# as elements within a regular set, allowing for a form of nesting.
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[156], line 3
      1 # Can a set be nested? Explain.
----> 3 s={5,3,2,{'dress','hat'},7}
      4 print(s)

TypeError: unhashable type: 'set'
```

In [157…
```python
# What are some real-world use cases for sets in Python?

# The removal of duplicates by using sets is a real-world usecase for sets in python.
# We can find differences between the data using sets in python.
```

Python Questions on Tuple and Set

In [181…
```python
# Write a Python program to find the repeated items of a tuple.
tp = (5, 6, 7, 3, 'cat' ,'bat', 556.7, 5, 'bat')

ln = []

for item in tp:
    if tp.count(item) > 1:
        ln.append(item)

e = set(ln)

print(tuple(e))
```

```
('bat', 5)
```

In [184…
```python
# Given a tuple of numbers, write a program to find the second largest element.
tp1 = (56,78,33,22,77)
tp1_srt = sorted(tp1)
tp1_srt
print(tp1_srt[-2])
```

```
77
```

In [186…
```python
# Write a program to convert a tuple of strings into a single string.
tp2 = ("h", "i")
string = ''.join(tp2)
print(string)
```

```
hi
```

In [187…
```python
# Write a program to remove duplicate elements from a list using a set.
lst = [2,2,2,2,5,5,6,6,3,2,6,7,7,2]
st2 = set(lst)
st2
```

Out[187…
```
{2, 3, 5, 6, 7}
```

In [190…
```python
# Given two sets, write a program to check if one set is a subset of another.
sb = {1 , 2 , 3}
sb1 = {2 , 3}
print(sb1.issubset(sb))
```

```
True
```

In [114…
```python
# Write a Python function that takes a tuple of numbers and returns a new tuple with only even numbers.
```

In [192…
```python
# Write a program to find all common elements between two sets.
s1={1,2,3,4,5,5}
s2={4,5,6}
print(s1.intersection(s2))
```

```
{4, 5}
```

In [116…
```python
# Write a program to count the frequency of each item in a tuple.
```

In [194…
```python
# Write a Python program to create a union, intersection, and difference of two sets entered by the user.
s = set(input())
s1 = set(input())
print(s.union(s1))
print(s.intersection(s1))
print(s.difference(s1))
```

```
{' ', '4', '1', '5', '3', ','}
{'3', '5', ',', '4'}
{'1', ' '}
```

In [121…
```python
# Given a tuple of mixed data types, write a program to extract all string values.
my_set = {56, 50235, 243, 367, 'Soap', 54.767}
a,b,c,d,e,f = my_set
print(a,b,c,d,e,f, sep = ", ")
```

```
243, 54.767, 56, 50235, Soap, 367
```

```
In [ ]:
```